TECHNICAL PRESENTATION

Lumelle AI

Local AI, no internet



Varvara Surnova

THE TASK

An assistant without the cloud



PROBLEM

Data in the cloud



GOAL

AI on the laptop



RESULT

Works offline

FEATURES

What Lumelle can do



Chat



Write



Translate



Slides



Learn



Images



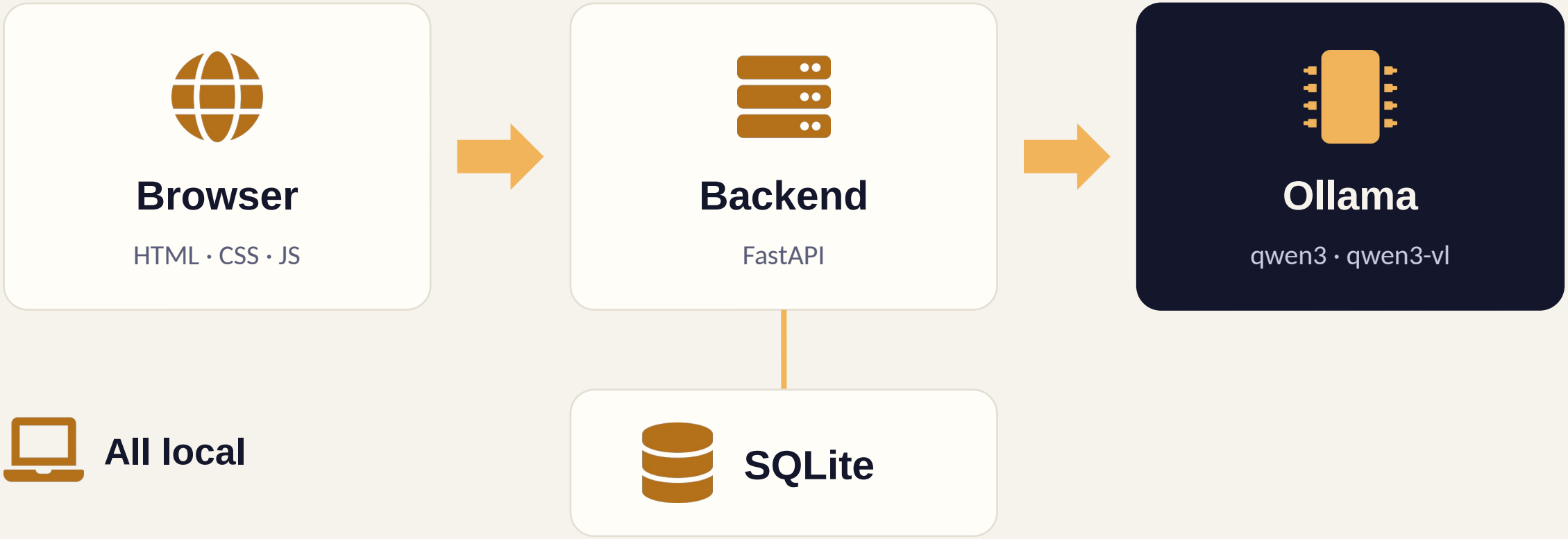
History



5 languages

ARCHITECTURE

Three layers, one computer



STACK

Technologies



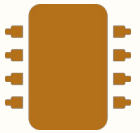
Qwen3 8B

text



Qwen3-VL 8B

images



Ollama

model runtime



FastAPI

backend · python-pptx



SQLite

history

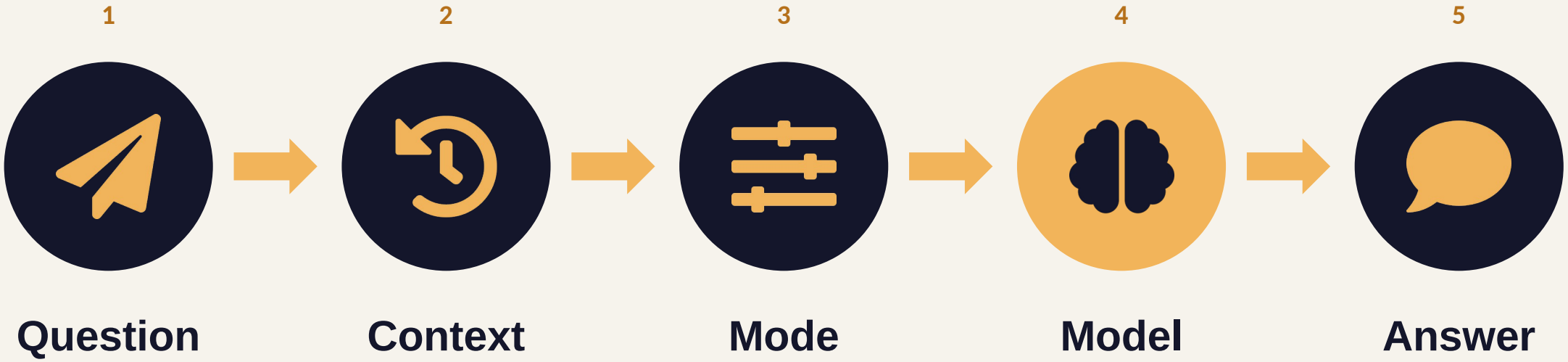


HTML · CSS · JS

interface

REQUEST FLOW

The path of one message



last 20 messages · mode prompt · qwen3 or qwen3-vl

MODEL SETTINGS

Temperature by mode

Chat	Write	Translate	Slides	Learn
0.4	0.7	0.1	0.5	0.4
precise	lively	exact	structured	balanced



`think` — reason before answering

DATA STORAGE

Two tables in SQLite

chats

- id
- title
- mode
- created_at

1:N

messages

- id
- chat_id → chats
- role
- content
- images
- created_at

REST API

8 endpoints

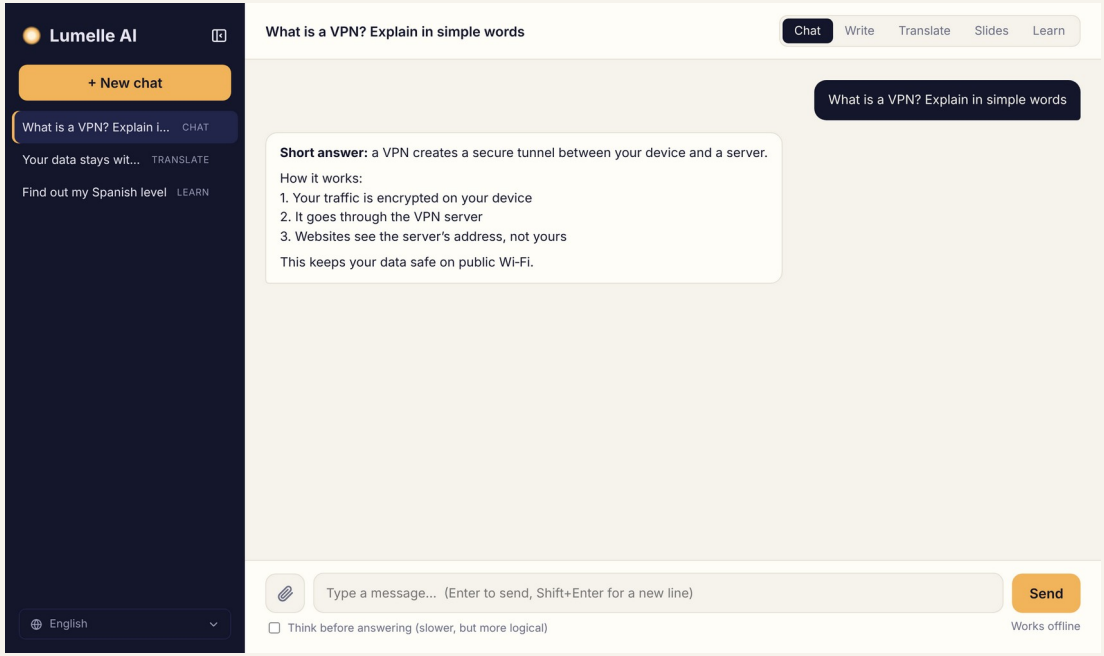
POST	/ask
POST	/chats/{id}/ask
POST	/chats
PATCH	/chats/{id}
GET	/chats
DELETE	/chats/{id}
GET	/chats/{id}
GET	/decks/{file}

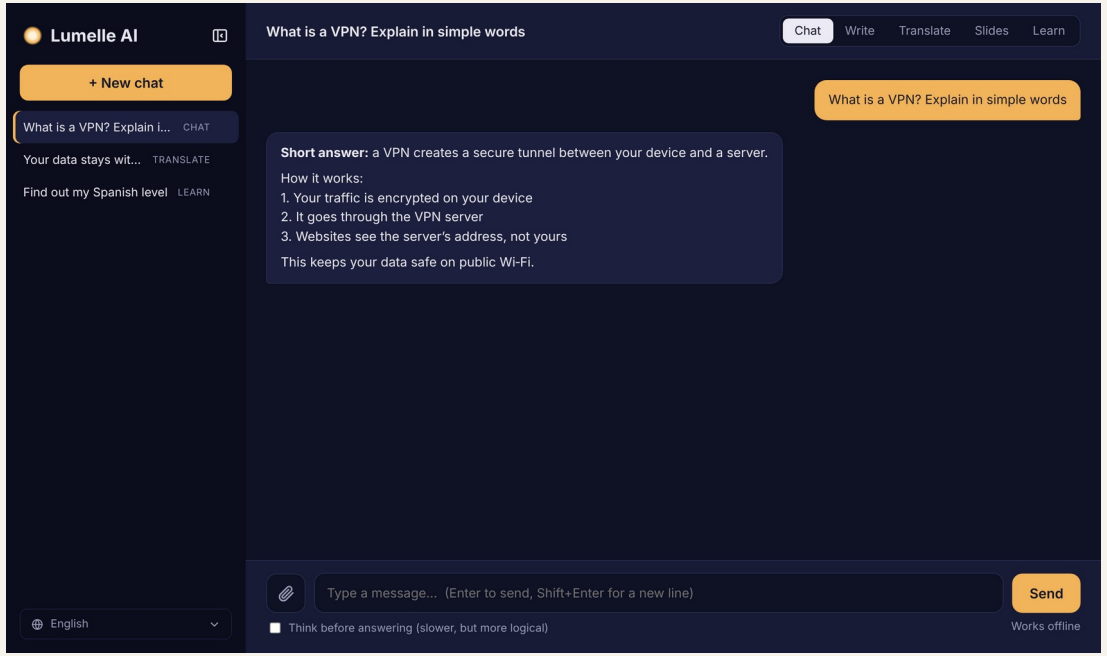


/docs

INTERFACE

Light and dark theme





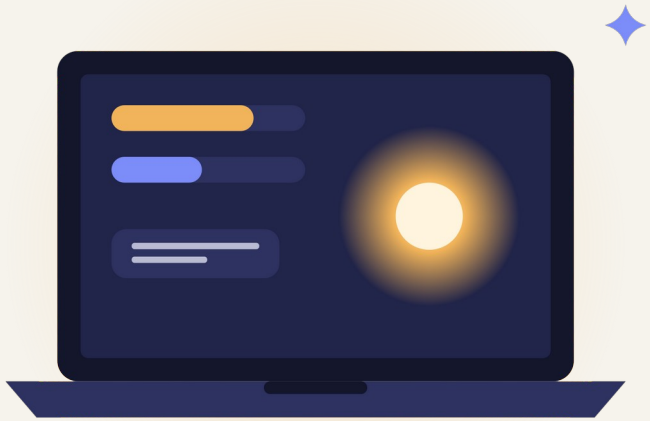
REQUIREMENTS

What it runs on

16 GB
RAM

11 GB
on disk

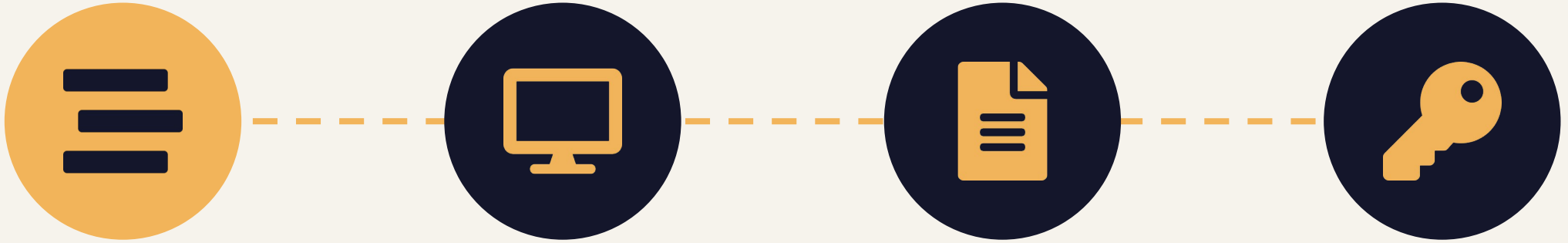
0
network requests



Windows · macOS · Linux

ROADMAP

Plans



Streaming

Desktop app

Your documents

Password



Thank you for your attention!

Any questions?